FLEET & SHORE PERSONNEL TRAINING

Ship Safety Officer

The role and duties of the ship safety officer, the tasks of the safety committee, and the Company risk assessment and management guidelines.

ISM Code

Code of Safe Working Practices

SSF-25

MPM System

COURSE MAP

What we will cover

01 **The Ship Safety Officer**
Appointment, role and duties on board

02 **Inspections and Reporting**
SSF-25, MPM and non-conformities

03 **The Safety Committee**
Composition, powers and tasks

04 **Risk Assessment Guidelines**
Objective, scope and components

05 **Risk Management**
Controls, approval and annual review

06 **The Assessment Process**
Stages, flow chart and questions

07 **Risk Estimation and Rating**
Severity, probability and action

SECTION 01

01

The Ship Safety Officer

The safety officer of the ship is entitled to look over all the duties related to the safety of the ship and its crew.

APPOINTMENT

Who holds the role on board

The role is a designated appointment with defined authority, held alongside the duties of Chief Officer.

Ship Safety Officer

The Chief Officer

- The Chief Officer is the vessel's designated Ship Safety Officer
- Inspects all areas of the vessel on a regular basis for safety compliance
- Reports any deficiencies noted with the SSF-25 form
- Available to advise those preparing and reviewing risk assessments

Training Officer

Also the Chief Officer

- Responsible to the Master for the proper training of all crew members
- Training of engine personnel is made in co-operation with the Chief Engineer
- Ensures adequate study periods and duties commensurate with such training
- Understudies the Master's duties so he may assume command in case of illness or accident

Evidence of competence

Held on file

- Evidence that the safety officer has undertaken an appropriate safety officer training course should be provided
- Trained in the subjects set out later in this section
- Familiar with the principles and practice of risk assessment

THE CORE OF THE ROLE

Look over every duty related to the safety of the ship

The safety officer is entitled to look over all the duties related to safety of the ship. The role should be a positive one, seeking to initiate or develop safety measures before an incident occurs rather than afterwards.

Before, not after

Every accessible area

Advise and escalate

Hazards, awareness and inspection

The first group of duties concerns knowing the ship, keeping standards high and inspecting on a fixed cycle.

Look into all potential hazards

Examine every potential hazard to the health and safety of the ship and its crew.

Maintain safety consciousness

Ensure the crew maintains a high standard of safety consciousness and knows all the important aspects of the ISM Code related to safety.

Enforce the Code of Safe Working Practices

Ensure the provisions of the code of safe working practices and the safety instructions, rules and guidance for the ship's safety and health are complied with.

Inspect at least every three months

Ensure safety inspection of the ship is carried out at least every three months, or more frequently if required.

Two inspection cycles

The minimum interval for a safety inspection of the ship is three months. The Company requirement goes further: the SSF-25 Environmental, Health and Safety Inspection is completed and reported monthly.

Advice, complaints and the safety management system

Support the safety committee

Help the ship safety committee to take substantial steps for enhancing the safety of the ship.

Take up crew complaints

Look into crew complaints related to health and safety.

Recommend hazard removal

Make recommendations to the ship's Master to remove a potential hazard which might lead to an accident or harm to the crew.

Build an efficient SMS

Help in making an efficient safety management system (SMS).

Report deficiencies to the Master

Provide information to the Master regarding deficiencies related to the ship's health, safety and security.

Feed the maintenance plan

Recommend important aspects related to safety in the ship's maintenance plan.

Investigation, records and stopping work

Investigate accidents properly

Properly investigate any accident involving the death of a crew member, and any major or minor accident.

Report non-compliance

Inform the Master when someone from the crew fails to work according to the regulations mentioned in the ISM Code.

Maintain accident records

Keep a record of all accidents taking place on the ship, including death, major or minor injury and near-death experiences, and make this information available to the Master, the safety representative or any official of the Company.

Stop unsafe operations

Stop any ship operation which might cause damage to the ship or harm to any of the ship's crew, inform the Master of the same and take necessary steps.

What the role includes, and what it does not

The safety officer identifies problems and escalates them. The role is deliberately kept separate from carrying out the repair work itself.

Within the role

- Inspecting all accessible areas of the vessel for safety compliance
- Identifying equipment deficiencies
- Investigating incidents, unless involved within and unless another person is assigned
- Advising those preparing and reviewing risk assessments
- Raising awareness and preventing accidents

Outside the role

- The function of the safety officer may not involve equipment maintenance, although it does include identifying equipment deficiencies
- Investigating an incident in which the safety officer was involved
- Investigating where another person has been assigned to do so

Committee, risk and supervision

The safety officer should be trained for the subjects below and is responsible for each of them.

The tasks of the safety committee
What the committee exists to do, and how the safety officer supports it.

The rights and roles of committee members
The rights and roles of members of the safety committee under the regulations.

Risk assessment and management
How to carry out risk assessment and management for the tasks performed on board.

Advising on safety concerns
How to provide the necessary advice to resolve safety concerns or problems, and to encourage adherence to prevention principles.

Supervision of safety tasks
Supervision of safety tasks assigned to crew and other seafarers on board, and passengers where applicable.

Advisory, not adversarial

The safety officer should be familiar with the principles and practice of risk assessment, and should be available to advise those preparing and reviewing risk assessments.

Investigation, information and communication

Investigation and analysis

Accident and incident investigation, analysis and making appropriate corrective and preventative recommendations to prevent their recurrence.

Obtaining information

How to obtain relevant information on a safe and healthy working environment from the competent authority and the Company.

Multinational crews

Effective means of communication with a multinational crew, if applicable.

Commitment

The commitment required to promote a safe working environment on board.

IN PRACTICE

How the safety officer deals with a hazard

The duties above combine into one repeatable sequence, whether the hazard is found during inspection or reported by a crew member.

1. Identify

Find the potential hazard to the health and safety of the ship and its crew.

2. Assess

Judge the significance of the hazard and whether existing safeguards control it.

3. Recommend

Make recommendations to the Master to remove the hazard before it causes an accident.

4. Stop if necessary

Stop any operation which might damage the ship or harm the crew, and inform the Master.

5. Record

Record the finding and the action taken, and make the record available to the Company.

Where the Chief Officer is a day worker

On vessels where the Chief Officer is a day worker, the following are required.

Bridge familiarisation

Fully familiarised with the bridge the soonest possible after joining. The familiarisation shall include at least four bridge watches.

Standing orders signed

The Master's Standing Orders are to be signed by the Chief Officer in addition to the Navigating Officers.

Standby and pilotage

At standby and under pilotage the Chief Officer's station shall be on the bridge, as far as possible.

Relieving the Master

Where prolonged standby such as a long river passage means the Master needs rest or must leave the bridge, the Chief Officer shall relieve him.

Bridge log book

The bridge log book requires the Chief Officer's signature, daily whilst attending the bridge, making himself fully aware of the passage, the day's run and any defects.

Passage plan

The passage plan is to be read, understood and signed by the watch Officer and also by the Chief Officer, even if he is on day work.

SECTION 02

Inspections and Reporting

02

One of the primary functions of the safety officer is to inspect all areas of the vessel on a regular basis for safety compliance and to report any deficiencies noted.

THE FORM

SSF-25 Environmental, Health and Safety Inspection

Company specific form SSF-25 is used for conducting and reporting of environmental, health and safety inspection.

What it covers

The inspection check list covers all accessible areas of the vessel and equipment. The safety officer inspects each accessible part of the vessel by following the SSF-25 form.

How often

The environmental, health and safety inspection is carried out monthly and the results are recorded. Deficiencies noted are reported with the SSF-25 form.

Where the report goes

The report is forwarded via the MPM system to the Company every month, with any non-conformances if identified.

Non-conformity reports

A non-conformity report will be issued via the MPM system for each finding raised during the inspection.

THE MONTHLY CYCLE

From inspection to closure

Every monthly inspection follows the same route, so that findings are visible ashore and can be tracked to closure.

1. Inspect

Each accessible part of the vessel and its equipment, following the SSF-25 check list.

2. Record

Results recorded on the SSF-25 Environmental Health and Safety Inspection Form.

3. Forward

Report sent to the Company via the MPM system every month.

4. Raise

A non-conformity report issued via MPM for each finding raised.

5. Close out

Corrective action tracked to closure with Company oversight.

PURPOSE

Why the inspections and the reports matter

1	3	Fleet
SSF-25 inspection and report every month	Months, the maximum interval between safety inspections	Wide view of recurring findings, built from ship reports

Raise awareness — The purpose is to raise awareness of the hazards present in the working spaces.

Prevent accidents — Findings are acted on before they can develop into an accident or an injury.

Identify regular occurrences — Identify regular occurrences that might require the Company’s intervention on a fleet-wide basis.

Who does what with an inspection finding

On board

- The safety officer conducts the monthly SSF-25 inspection and records the results
- Deficiencies are reported with the SSF-25 form
- The safety officer also investigates incidents, unless involved within and unless another person is assigned
- Records are made available to the Master, the safety representative and Company officials

Ashore

- The Company receives the monthly report via the MPM system
- A non-conformity report is issued via MPM for each finding
- The Company identifies regular occurrences that need intervention on a fleet-wide basis
- Evidence of the safety officer’s training course is held on file

THE MEASURE OF A GOOD SAFETY OFFICER

Initiate safety measures before an incident, not afterwards

The safety officer’s role should be a positive one, seeking to initiate or develop safety measures before an incident occurs rather than afterwards.

Proactive

Advisory

Evidence based

SECTION 03

03

The Safety Committee

A committee formed on board under the chairmanship of the Master, working with the sole goal of enhancing the safety standard on board ships.

COMPOSITION

How the committee is formed

In order to ensure that the ship and its crew follow all safety procedures while doing work and maintain a safe working environment, a safety committee is formed on ships.

Chaired by the Master

The safety committee is formed under the chairmanship of the Master.

The safety officer

The Chief Officer, as designated Ship Safety Officer, sits on the committee.

The safety representative

The crew safety representative sits alongside the safety officer.

Other competent persons

The committee comprises other competent persons as the ship's operation requires.

Additional members

More crew members can be included in the committee if the need arises.

A single goal

The committee works with the sole goal of enhancing the safety standard on board, by ensuring all safety procedures and practices are followed by the crew.

Standards, attitude and representation

The main tasks of the safety committee begin with protecting the standard of safe working practice on board.

Protect safe working practices
Ensure that safe working practices and standards are followed on the ship and are not compromised at any cost.

Build a safety-first attitude
Improve the standards of safety by enhancing a safety-first attitude among crew members.

Recommend improvements
Make recommendations regarding enhancement of occupational health and safety measures on ships.

Represent the crew
Act as the representative of the crew to address concerns and queries to the ship management.

Inspect the records
Inspect the safety officer’s records.

Policy, records, equipment and meetings

Act on policy

Take appropriate actions pertaining to occupational health and safety policies.

Record the meetings

Keep a record of safety meetings, suggestions, progress and actions taken. All records of the committee are properly noted down in the official log book.

Provide tools and publications

Ensure that necessary safety tools and equipment are available to the crew members, along with safety publications.

Check accident reports

Look into the accuracy of accident reports.

Meet every month

Make sure that safety meetings are held every month, or whenever the need arises.

Same powers as representatives

The committee has the same powers under the regulations as those possessed by safety representatives.

AUTHORITY

The committee has the same powers as safety representatives

Safety committee is an important body on board ships. It has the same powers under the regulations as those possessed by safety representatives.

Chaired by the Master

Meets monthly

Recorded in the official log book

SECTION 04

04

Risk Assessment Guidelines

RAS is a systematic way of gathering, evaluating and recording information leading to recommendations for safeguards, in response to an identified hazard.

VOCABULARY

The terms used throughout this section

Six abbreviations appear repeatedly in the Company procedures covered here.

RAS

Risk assessment. A systematic way of gathering, evaluating and recording information leading to recommendations for safeguards and action, in response to an identified hazard.

SMS

Safety Management System. The risk assessment guidelines are to be read in conjunction with the SMS Manuals.

MPM

The planned maintenance system. All risk assessments and the generic RAS catalogue are held within MPM.

SSF-25

The Company form for conducting and reporting the environmental, health and safety inspection.

DPA

Designated Person Ashore. Reviews preventive measures and risk assessments where no safe working procedure exists.

ALARP

As low as reasonably practicable. The level to which the risks associated with a task must be reduced.

SOURCES

What a risk assessment is compiled from

Reference sources are referred to when compiling a risk assessment. The assessment is not written from memory.

Industry organisations

Reference sources from industry organisations for the operation concerned.

Code of Safe Working Practices

The Code of Safe Working Practices for Merchant Seafarers.

IMO guidelines

Applicable IMO guidelines.

Legislation

Legislation in force for the vessel and the operation.

Start from the catalogue

A catalogue of generic risk assessments is available within MPM System - HSEQ - Definitions - RAS - RAS Template section. Use it as the starting point, then add the specific circumstances of the task.

Why the Company operates a risk assessment system

Objective

- Understand risks involved in the management of the Company
- Cover routine, non-routine or unfamiliar, and hazardous operations which pose a risk to human, environment, assets and reputation
- The RAS must become a learned behaviour for all levels of the Company
- The Company institutes and maintains a risk-decision framework and culture for managed vessels

Scope

- Provide an overview of the components of handling unfamiliar operations or tasks on the managed vessels
- These components comprise the Safe System of Work, based on a risk assessment approach
- To be read in conjunction with the SMS Manuals
- Outlines the requirements for carrying out unfamiliar and risky tasks

What the procedure is designed to guarantee

Tasks are planned in a systematic manner so that risks are established in terms of consequence and likelihood, and safeguards are in place before work starts.

1. Plan systematically

The task is planned in a systematic manner rather than improvised at the work site.

2. Identify hazards and people

The risk assessment identifies all hazards associated with a task and any personnel at risk.

3. Evaluate the risk

Risks are established in terms of consequence and likelihood, and the risks to be taken are evaluated.

4. Specify safeguards

Appropriate safeguards, the control measures, are specified for each hazard identified.

5. Then commence

Safeguards are put in place prior to allowing any unfamiliar and potentially risky works to commence.

POTENTIAL RISKS

The five areas a hazard can damage

Every assessment considers the consequence of the hazard against each of these five areas.

Human life

Injury, ill health or fatality to any person involved.

Environment

Loss of containment, spill or release to the marine environment.

Property

Damage to the vessel, its equipment or third-party assets.

Operational loss

Missed voyages, delay, additional work and financial loss.

Reputation

Impact on the Company, from limited to major international.

Risk analysis covers several types of analysis

The risk analysis term covers several types of analyses that will all assess causes for accidents and consequences of accidental events. These are examples of the simpler analyses.

Technique	What it examines
Safe Job Analysis	Takes a single job step by step to identify the hazard at each step and the safeguard required.
FMECA	Failure Modes, Effect and Critical Analysis. How a component can fail, what follows from the failure and how critical it is.
Preliminary Hazard Analysis	An early, broad screening of hazards, used to decide what needs deeper study.
HAZOP	Hazard and Operational Studies. Structured examination of deviations from the design intent of an operation.

The technique is chosen to suit the task being assessed. Teamwork is required in every case.

COMPONENTS

The six components of risk assessment

1. **Initiation**
   The decision to assess, triggered by a new, non-routine, unplanned or unfamiliar task.

2. **Hazard identification**
   Establish the hazards present at the place of work and the significant risks arising out of the work activity.

3. **Risk assessment**
   Probability, consequences and uncertainty.

4. **Risk management**
   Efficacy, feasibility and impacts of the controls selected.

5. **Risk interrelation and communication**
   How the risks relate to one another, and how they are communicated to those at risk.

6. **Policies and procedures**
   Policies and procedures relating to risk, which the assessment must respect and can improve.

OUTPUTS

The questions the RAS gives a structure to solve

The value of a risk assessment lies in the answers to these six questions. A form completed without answering them adds nothing to the safety of the task.

What can be done to eliminate or reduce the hazard?
The recommendations for safeguards to be put in place, and the action to be taken in response to the identified hazard.

How relevant are the options?
Whether each option actually addresses the hazard that has been identified.

How feasible are the options?
Whether the option can be implemented on board with the resources available.

What impact would any options have?
The impacts on human life, environment, property, operational loss and reputation.

What is the level and type of uncertainty?
Uncertainty is one of the components of risk assessment, alongside probability and consequences.

What is the best option, that is, what can we do?
The chosen course of action, recorded with the safeguards to be implemented before work starts.

BENEFITS

What risk assessment delivers

The benefits of risk assessment are measurable, in safety terms and commercially.

Reduce overall costs

Fewer losses to absorb across the fleet.

Reduce loss of life and injuries

The primary purpose of the whole process.

Reduce damage

Less physical damage to vessel and equipment.

Reduce liability

Less exposure to legal and contractual claims.

Reduce exposure

Lower exposure for both the ship and the Company.

Develop performance and quality

Findings feed back into the operating procedures.

Protect reputation

Protect the standing of the Company.

Improve safety culture

Shared assessments spread best practice fleet wide.

SECTION 05

Risk Management

A continuous, systematic process of identifying and controlling risks in the range of operations which could cause harm.

A continuous, systematic process

The Company has introduced a continuous, systematic process of identifying and controlling risks, so that decisions can be made as to whether enough precautions have been taken or whether more should be done to prevent harm.

The aim
Minimise accidents, operational loss and ill health on board ship, and manage operational risks fleet wise.

Targets for close out
Targets are set for close out of preventive measures identified during the assessments.

An active file ashore
The office keeps an active file, investigates any delays and expedites closure.

Controls into procedures
All risk mitigation measures identified for hazards are incorporated into the safe working procedures and implemented prior to commencement of work.

Response elements
The process includes response elements to limit the impact of any unplanned occurrence, detailed on the Risk Assessment Form.

Permit to work and final approval

The process will ensure the permit to work system for both planned and unplanned tasks is used.

1. Risk assessment

The assessment is carried out for the task and the safeguards are identified.

2. Permit to work

The permit to work system is used, for planned and unplanned tasks alike.

3. Confirm implementation

The risk assessment form is in use to confirm implementation of the measures.

4. Final approval

Final approval for commencement of work is subject to implementation of mitigation measures.

When the residual risk is still too high

Consider alternatives

- Preventive measures and alternative methods of work are considered to ensure safe completion of the work
- They are documented in the risk assessment process
- This applies where the residual risk has been determined to be unacceptable

Do not start

- Risk assessments whose final factor is calculated as intolerable are forwarded immediately to the Company for review and approval
- The job will not start until approval is received from the Company
- If the risk factor is intolerable, work shall not be started until the risk is reduced

Who reviews and approves

- Measures taken for risk minimisation as a preventive action are highlighted in the risk assessment
- They are reviewed by the DPA, to improve the operating procedures included in the Safety Management System
- Where no safe working procedure exists, the assessment is reviewed by the DPA and must be approved by the Fleet Manager

Where risk assessments are held and shared

The Company collates all risk assessments within planned maintenance system MPM for best practice sharing, in order to improve the company safety culture.

One database
Risk assessments are kept within the database of MPM, so ship and shore work from the same records.

Generic RAS catalogue
A catalogue of generic risk assessments is available within MPM System - HSEQ - Definitions - RAS - RAS Template section.

Consistent standards
The Company reviews and collates all vessel-generated and shore-generated risk assessments to check the standards are consistent.

Applied across the fleet
Common risk assessments are applied across the fleet so best practices for common areas are shared, and Company procedures are amended if required.

Changed with the crew
Any alterations to risk assessments coming from a review should be made in consultation with ship crew.

What the annual review has to consider

Risk assessments and templates, including the catalogue, are reviewed annually for validity and to ensure they remain fit for purpose.

- Risk assessments and templates, including the catalogue, are reviewed annually for validity
- The effect of new legislation and/or equipment is incorporated into the risk assessment
- The changes in manning level or levels are taken into account
- The non-routine tasks are considered, and may become standard tasks following review
- The annual review process ensures that all risk assessments remain relevant
- Common risk assessments are checked for consistent application across the fleet
- Best practices for common areas are shared across the fleet where required
- Company procedures are amended where the review shows this is required
- Any alterations coming from the review are made in consultation with ship crew, and records are maintained

Living documents

A risk assessment that has not been reviewed against new equipment, new legislation or a changed manning level is no longer an assessment of the risk actually being run.

AFTER THE REVIEW

Familiarisation and availability

A review only takes effect when the people who have to use the risk assessment know what has changed and can find the current version.

Everyone is familiarised

All relevant shore and ship personnel are familiarised with the content and review results of the risk assessments.

Records are kept

The familiarisation records are maintained on board or ashore.

Assessments are available

All approved and reviewed risk assessments are available to all relevant personnel within planned maintenance system MPM.

DAILY USE

Bringing risk management into daily work

In order to incorporate risk management concepts into daily operational, maintenance and support activities, the Company's Risk Assessment forms should be used.

Use the Company forms

The Company's Risk Assessment forms are used for operational, maintenance and support activities.

Add what is specific to the task

If any additional specific circumstances arise, or hazards are identified, they should also be considered during the risk assessment.

Hazards first, then risks

The assessment should first establish the hazards present at the place of work, then identify the significant risks arising out of the work activity.

New, non-routine and unplanned

The process includes assessing new, non-routine and unplanned tasks, detecting hazards, assessing risks including those to health and hygiene, and implementing and monitoring risk controls to support risk-based decision-making.

EXISTING PRECAUTIONS

Controls the assessment must take into account

The assessment should take into account any existing precautions that control the risk.

Permits to work

A valid permit for the space or the task.

Restricted access

Access limited to authorised persons.

Warning signs

Use of warning signs at the place of work.

Agreed procedures

Procedures already agreed for the operation.

Protective equipment

Personal protective equipment for the hazard.

THE STANDARD TO MEET

As low as reasonably practicable

In summary, the risk assessment should ensure that protective and precautionary measures are taken which will reduce the risks associated with a task to a level that is considered to be as low as reasonably practicable.

ALARP

Documented

Implemented before work starts

06

SECTION 06

The Assessment Process

Process chart of risk assessment and risk estimation, summarising hazard identification, risk assessment and control that contribute towards a safe system of work.

PURPOSE

What the process chart is for

The objective of this procedure is to summarise the processes associated with hazard identification, risk assessment and control that contribute towards achieving a safe system of work.

The Master collects the information

The Master collects all information that he has and completes a risk assessment before the task is commenced.

Help for the people on board

This procedure aims to help persons on board to assess risks in the workplace.

Stop and think first

The process has two stages, each dependent on stopping and thinking about the task before it commences.

Before the task, not during it

The assessment is completed before the task is commenced. Once the work has started, the only remaining option is to stop it.

TWO STAGES

Stage 1 and Stage 2 of the assessment

Stage 1

A broad overview

- Carried out by the person appointed to carry out the work
- Determines whether the hazards are significant
- If they are, determines whether the risks can be controlled by existing means
- Existing means include safeguards within a planned maintenance routine, established or local procedures, and permit to work

Stage 2

A formal assessment

- A formal assessment of the task
- Required when the permitting authority, the senior person overseeing the task, judges that additional safeguards will be needed
- Aimed at minimising the risks of being hurt or causing harm within the working environment

From problem to execution

1. **Problem**
   A task, hazard or situation is identified that needs to be assessed.

2. **Estimate accident severity**
   Estimate how bad the consequence would be, using severity categories 1 to 5.

3. **Estimate accident likelihood**
   Estimate how likely the accident is, using the probability scale 1 to 5.

4. **Risk determination**
   Severity multiplied by likelihood gives the risk factor and its rating band.

5. **Is risk acceptable?**
   If no, go to solution: apply further controls and assess again. If yes, proceed.

6. **Execution**
   The task is carried out with the safeguards implemented and the permit in place.

FIVE QUESTIONS

The questions to answer when carrying out a risk assessment

The types of question that should be answered when carrying out a risk assessment are as follows.

What can go wrong?

An identification of the hazards and accident scenarios, together with potential causes and outcomes.

How bad and how likely?

An evaluation of the risk factors, the consequence and the probability.

Can matters be improved?

An identification of risk control options to reduce the identified risks.

What is the effort, and how much better would it be?

A determination of the benefit and effectiveness of each risk control option.

What action should be taken?

An identification of the appropriate course of action to deliver a safe activity, based on the hazards, their associated risks and the effectiveness of alternative risk control options.

Where it is recorded

The intended operation and work are listed at the file of sample RAS forms.

SECTION 07

07

Risk Estimation and Rating

Consequence and probability are each scored from 1 to 5. Their product is the risk factor, and the risk factor decides what happens next.

THE SCALE AT A GLANCE

How a risk factor is built

5
Severity categories, from negligible to catastrophic

5
Probability levels, from not credible to probable

25
Highest possible risk factor, five by five

- **Tolerable, 1 to 6** — No additional controls are necessary; the risk is already ALARP. Monitoring is required.
- **Moderate and substantial, 8 to 12** — Efforts should be made to further reduce risks wherever possible before work starts.
- **Intolerable, 15 to 25** — Work must not be started until the risk has been reduced.

Severity: people and assets

Cat	Definition	People	Assets
1	Negligible	Injury is not credible (First Aid Case)	Less than 1,000$
2	Minor	Only a minor injury is credible (Medical Treatment Injury)	Less than 10,000$ but greater than 1,000$
3	Significant	A single serious injury is credible (Restricted Work Case, RWC)	Greater than 10,000$ but less than 99,999$
4	Critical	Fatality or multiple serious injury is credible (Serious Day Away from Work Case, DAFWC)	Vessel disabled, or 100,000$ and above
5	Catastrophic	Multiple fatality is credible (Fatality)	Explosion, collision, grounding or damage greater than 100,000$

Severity: operational loss, environment and reputation

Cat	Operational and financial loss	Environment	Reputation
1	No missed voyage	Loss of containment less than 1 Bbl, not on deck or to environment	Zero impact
2	Less than 10K$, additional work, no missed voyage	Environmental impact not existent	Limited impact
3	Longer operational and financial loss, 10K$ to 100K$	Local environmental impact	Considerable impact
4	Major operational and financial loss, 100K$ to 1M$, missed voyages	Regional environmental impact	Major national impact
5	Greater than 1M$	Major spill of 100 Bbls or more, or uncontrolled gas release over 10 tonnes or volumetric equivalent	Major international impact

SCORING

Estimating a risk in four steps

Risk estimation is deliberately simple, so that it can be done consistently by different people on different ships.

1. Score the severity

Choose the category 1 to 5 for the worst credible consequence to people, assets, operations, environment or reputation.

2. Score the probability

Choose the level 1 to 5 against experience on this vessel, in the Company and in the shipping industry.

3. Multiply

Severity multiplied by probability gives the risk factor, between 1 and 25.

4. Act on the band

Compare the factor with the rating band and take the action that band requires.

The probability, or likelihood, scale

Probability is scored from 1 to 5 against experience on this vessel, in our Company and in the shipping industry.

Score	Definition	Test to apply
1	Not credible	Never heard of in the shipping industry.
2	Unlikely	Heard of in the shipping industry.
3	Possible	Similar incident has occurred in our Company, and/or heard several times in the shipping industry.
4	Likely	Happens several times a year in our Company.
5	Probable	Happens several times a year on this vessel.

Severity multiplied by probability

Read the severity category down the left and the probability across the top. The cell value is the risk factor.

Severity	(1) Not credible	(2) Unlikely	(3) Possible	(4) Likely	(5) Probable
1 Negligible	1	2	3	4	5
2 Minor	2	4	6	8	10
3 Significant	3	6	9	12	15
4 Critical	4	8	12	16	20
5 Catastrophic	5	10	15	20	25

Increasing probability runs from left to right; increasing consequence runs from top to bottom.

What the risk factor requires you to do

Tolerable

1, 2, 3, 4, 5, 6

- No additional controls are necessary
- The risk is already as low as reasonably practicable, ALARP
- Monitoring is required to ensure that the controls are maintained

Moderate and substantial

8, 9, 10, 12

- Efforts should be made to further reduce risks wherever possible before work starts
- Where consequences are extremely harmful, further assessments may be necessary
- Those assessments establish more precisely the likelihood of harm, as a basis for determining the need for improved control measures

Intolerable

15, 16, 20, 25

- Work must not be started until the risk has been reduced
- Where risks involve work in progress, urgent actions should be taken
- The assessment is forwarded immediately to the Company for review and approval

THE SAME JUDGEMENT, SIMPLIFIED

Reading the matrix

	Low	Medium	High
High	Medium	High	High
Medium	Low	Medium	High
Low	Low	Low	Medium

LIKELIHOOD

CONSEQUENCE →

Two questions, one decision

- How bad would it be? That is the consequence score
- How likely is it? That is the probability score
- Their product gives the risk factor and its rating band
- Tolerable: monitor and maintain the controls
- Moderate and substantial: reduce further before work starts
- Intolerable: do not start until the risk is reduced

ESCALATION

When a risk assessment must go ashore

Some assessments cannot be closed out on board. These are the rules for escalation and approval.

Send it ashore and wait

- Carry out a risk assessment for new, non-routine and unplanned jobs
- Forward immediately to the Company any risk assessment whose final factor is calculated as intolerable, for review and approval
- Hold the job until approval is received from the Company
- Highlight the measures taken for risk minimisation as a preventive action in the risk assessment
- Have those measures reviewed by the DPA, so they can improve the operating procedures in the Safety Management System
- Where no safe working procedure exists, have the assessment reviewed by the DPA and approved by the Fleet Manager

Never do this

- Start a job with an intolerable risk factor because the schedule is tight
- Treat a Stage 1 overview as a substitute for a formal Stage 2 assessment
- Use a generic template without considering the specific circumstances of the task
- Leave alternative methods of work undocumented where the residual risk is unacceptable
- Sign a permit to work before the mitigation measures are actually in place

THE FINAL RULE

If the risk factor is intolerable, work shall not be started

Work must not be started until the risk has been reduced. Where risks involve work in progress, urgent actions should be taken and the Company informed immediately.

Stop

Reduce the risk

Then start

Act before the incident, not after it

Inspect on the cycle, report through MPM, assess before the task commences, and stop the job when the risk factor says so.

SSF-25 inspection every month

Report findings through MPM

Assess before work commences

Intolerable means do not start